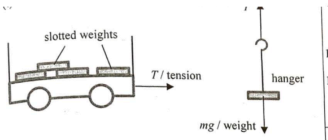
	Solution	Marks	Remarks
1.	(a) $Pt = ml$		
	$l = \dfrac{150 \times (5\times60)}{0.016}$	1M	
	$= 2812500\ \text{J kg}^{-1} \approx 2810\ \text{kJ kg}^{-1}$	1A	
		2	
	(b) $C\,(100-22) = m_w c_w\,(22-20)$		
	$C = \dfrac{0.100 \times 4200(22-20)}{(100-22)}$	1M	
	$= 10.76923\ \text{J °C}^{-1} \approx 10.8\ \text{J °C}^{-1}$	1A	
		2	
	(c) Since energy transferred by the boiling water is taken to be energy from the metal sphere / extra energy is transferred (by the boiling water) to the cup of water / the final temperature is higher.	1A	
	The true value of C is smaller (than the calculated value).	1A	
		2	
	(d) Not justified (with correct explanation).	1A	
	Copper – (good) conductor while polystyrene – (good) insulator or poor conductor, more energy will be lost to the surroundings by conduction via a copper cup / most part of the polystyrene cup is still at room temperature, negligible energy is absorbed.	1A	
		2	

Q2

Solution	Marks	Remarks
2. (a) $v = \frac{30 \times 2}{0.04}$	1M	
$= 1500\ \text{m s}^{-1}$	1A	
	2	
(b) (i) $\frac{p_1}{T_1} = \frac{p_2}{T_2}$ $\frac{18.5}{273+27} = \frac{p_2}{273+20}$	1M	
$p_2 = 18.068333\ \text{atm} \approx 18.1\ \text{atm}$		
	1	→ kinetic theory
(ii) As the temperature decreases, the speed / kinetic energy of the gas molecules decreases.	1A	
They collide less frequently and/or violently with the wall of the container / cylinder.	1A	
Thus, the pressure decreases.		
	2	
(c) (i) $n_0 = n_1 + n_2$ $p_0V_0 = p_1V_1 + p_2V_2$ fixed volume (container)	1M	Assume no leakage
$18.1 \times 0.012 = p_1 \times 0.012 + 4.0 \times 0.015$ $p_1 = 13.1\ \text{atm}$ (i.e. $\Delta p = 5.0$ atm)	1M	
	2	
(ii) The pressure decreases by 5.0 atm for inflating one balloon, assume k balloons can be inflated completely: $18.1 - 5.0k \geq 4.0$ or $k \leq \frac{18.1-4.0}{5.0} = 2.82$	1M	Accept: 18.1 atm → 13.1 atm → 8.1 atm $\therefore k = 2$
$\therefore k = 2$ ↳ seabed atm, 出唔到氣	1A	
Or Let V = volume of compressed gas under 4.0 atm $18.1 \times 0.012 = 4.0 \times V$ $V = 0.0543\ \text{m}^3$ $0.0543 - 0.015 \times k \geq 0.012$	1M	
$k \leq 2.82$ $\therefore k = 2$	1A	
	2	

	Marks		Remarks
3. (a) (i) Work done = K.E. gained $= \frac{1}{2}(0.22)(13.4)^2$ $= 19.7516\ J \approx 19.8\ J$	1M 1A	2	
(ii) By conservation of mechanical energy, $\frac{1}{2}mv^2 - \frac{1}{2}mu^2 = mgh$ $v^2 - 13.4^2 = 2(9.81)(1.0)$	1M		
$v = 14.113114\ m\ s^{-1} \approx 14.1\ m\ s^{-1}$	1A		For $g = 10\ m\ s^{-2}$, $v = 14.13\ m\ s^{-1}$
Or $v_y^2 = u_y^2 + 2as$ $v_y^2 = (13.4 \sin 55°)^2 + 2(-9.81)(-1.0)$ $v_y = 11.836662\ m\ s^{-1} \approx 11.8\ m\ s^{-1}$	1M		
speed $v = \sqrt{v_x^2 + v_y^2}$ $= \sqrt{11.8^2 + (13.4 \cos 55°)^2}$ $= 14.113079\ m\ s^{-1} \approx 14.1\ m\ s^{-1}$	1A	2	Accept $14.1\ m\ s^{-1}$ to $14.2\ m\ s^{-1}$
(b) (i) descending (a pointing downward) $-g$	1A	1	
(ii) $\frac{18}{2} = 13.4 \cos 55° \times t$ (constant horizontal speed) $t = 1.170971\ s \approx 1.17\ s$	1M 1A	2	
(c) Justified (with correct explanation) Horizontal component of the (initial) velocity is (greatly) increased (for a similar speed but at a much smaller angle (35° << 55°)) Or Vertical component of the (initial) velocity is (greatly) reduced / vertical height reached is smaller. Time of flight (to the highest point as well as) for the whole journey would be shortened.	1A 1A	2	
(d) (Wood is soft compared to concrete, thus) it lengthens the time of impact when the feet land on the ground. This reduces the impact force. [as net force $F = \frac{mv - mu}{t} = \frac{0 - mu}{t}$]	1A 1A	2	

4. (a) (i)



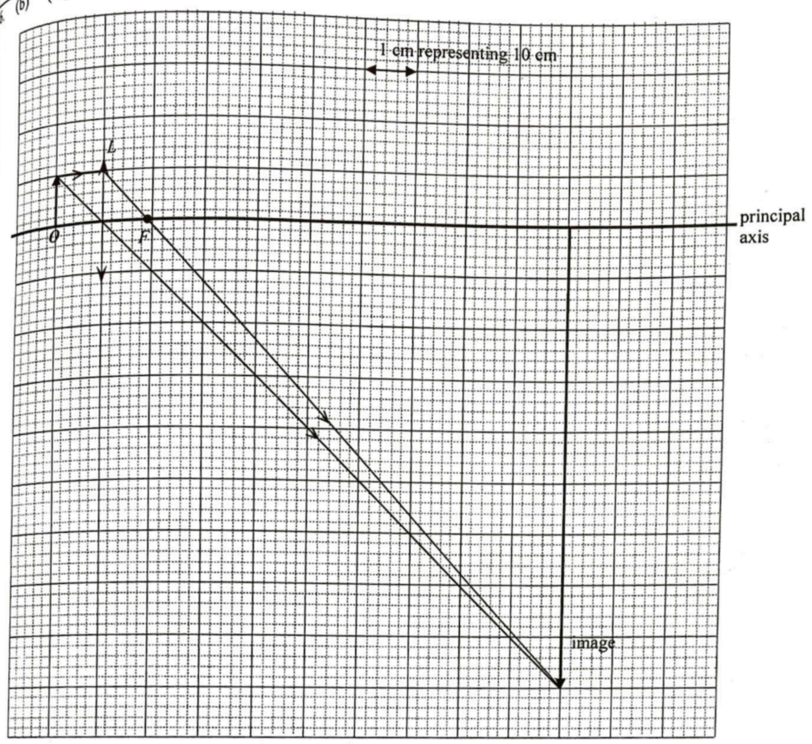
	Marks		Remarks
(a) (i) (diagram above)	1A 1A	2	
(ii) Smaller than (the weight of hanger / mg), i.e. $T < mg$, therefore by Newton's second law, a net force F is provided for the downward acceleration of the hanging mass Or therefore by Newton's second law, a downward net force F is provided for the acceleration of the hanging mass (both F and a are downward).	1A 1A	2	
(iii) Applying Newton's second law to the whole system, $mg = (M + 0.1)a$	1A	1	
(b) slope of the graph $= \frac{2.5}{0.1} = 25\ (m\ s^{-2}\ kg^{-1})$ or $0.025\ (m\ s^{-2}\ g^{-1})$ $mg = (M + 0.1)a$ $a = \frac{g}{M+0.1} \times m$ $\therefore$ slope of the graph $= 25 = \frac{9.81}{M+0.1}$ $M = 0.2924\ kg \approx 292\ g \approx 0.3\ kg$	1A 1M 1A	3	For $g = 10\ m\ s^{-2}$, $M = 0.3\ kg$

	Marks		Remarks
5. (a) (i) $F = \frac{\Delta p}{\Delta t} = \frac{2.60 \times 10^3 \times v}{1} = 5.20 \times 10^6$ $v = 2000\ m\ s^{-1}$	1M 1A	2	force on gas = thrust on rocket in magnitude
(ii) $F - mg = ma$ $a = \frac{F}{m} - g = \frac{5.2 \times 10^6}{3.6 \times 10^5} - 8.56$ $= 5.884444\ m\ s^{-2} \approx 5.88\ m\ s^{-2}$	1M 1A	2	Accept $5.88\ m\ s^{-2}$ to $5.90\ m\ s^{-2}$
(iii) The acceleration would increase. $F=ma$ Although the thrust remains the same, the mass of the rocket and/or g decreases.	1A 1A	2	
(b) (i) 24 hours / 1 day / 86400 s	1A	1	
(ii) $m\omega^2 r = \frac{GMm}{r^2}$ OR $\frac{mv^2}{r} = \frac{GMm}{r^2}$ OR $gR^2 = v^2 r = (\frac{2\pi r}{T})^2 r$	1M		

Q6

6. (a) (i) $f = \frac{c}{\lambda}$		
$= \frac{3\times10^{8}}{675\times10^{-9}}$	1M	
$= 4.444444 \times 10^{14}$ Hz $\approx 4.44 \times 10^{14}$ Hz	1A	
		2
(ii) $\frac{\sin 30°}{\sin\theta} = \frac{c}{v} = \frac{\lambda}{\lambda'}$		
$= \frac{675}{450}$	1M	
$\sin\theta = \left(\frac{450}{675}\right)\sin 30°$		
$\theta = 19.471° \approx 19.5°$	1A	
Definition: $n = \frac{\sin\theta_{air}}{\sin\theta'}$		2
(iii) The refractive index of glass for blue light is greater (than that for red light).	1A	
		1
(b) (i) real and/or inverted	1A	
		1
(ii) 10 cm	1A	
		1

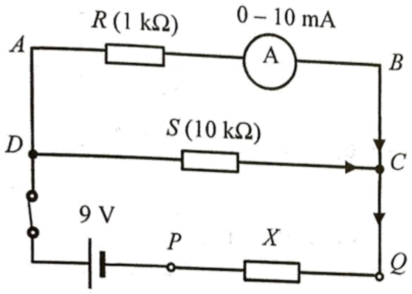
(b) (iii)



Location of F	1M		
Correct light rays from object to image	2M		
Correct position and size of image	1A		
Focal length of L = 9 cm	1A		Accept 8 cm to 10 cm
		5	
(iv) White light composes of different colours. Since the refractive indices of glass (the medium of the lens) are different for different colours,	1A		
the sizes of the colour images on the screen are slightly different. When these images overlap on the screen, colour edges are seen.	1A		
		2	

Q7

7.	(a)	(i)	Eddy currents are induced to oppose the change of magnetic flux due to the movement of the metal sheet.	1A	
			Moving to the left → accept magnetic field	1A	
				2	
		(ii)	Kinetic energy → Electrical energy → Internal (heat/thermal) energy	2A	
				2	
		(iii)	Limitation: Eddy braking only works when the vehicle is moving. e.g. The vehicle cannot park in stationary positions / still on a slope.	1A	
				1	
	(b)		Electrical energy consumed = 2 kW × $\frac{15}{60}$ h = 0.5 kW h	1M	
			Cost = \$ 1.1 × 0.5 = \$ 0.55	1A	
				2	
	(c)		Lamination, i.e. the cores of these devices are made up of multiple insulated sheets of metal.	1A	
				1	
	(d)		The magnetic field (produced by eddy currents) measured will decrease (where there are irregularities).	1A	
			It is because the crack will reduce the eddy currents.	1A	
				2	

	Marks	Remarks
8. (a) (i) $V_{AB} = (8.5 \times 10^{-3}\text{ A}) \times (1\text{ k}\Omega)$ $= 8.5\text{ V}$	1M 1A	
	2	
(ii) Current through $S = \dfrac{8.5\text{ V}}{10\text{ k}\Omega}$ $= 8.50 \times 10^{-4}\text{ A} = 0.85\text{ mA}$	1M 1A	As AB and CD are connected in parallel, current through S $= (8.5 \times 10^{-3}\text{ A}) \times \dfrac{1\text{ k}\Omega}{10\text{ k}\Omega}$
	2	
(iii) 	2A	
	2	
(iv) p.d. across $X = 9 - V_{AB} = 9 - 8.5 = 0.5\text{ V}$ Current through $X = (8.5 \times 10^{-3}) + (8.5 \times 10^{-4})$ $= 9.35 \times 10^{-3}\text{ A}$ Resistance of $X = \dfrac{0.5\text{ V}}{9.35 \times 10^{-3}\text{ A}}$ $= 53.476\ \Omega \approx 53.5\ \Omega$	1A 1M 1A	
	3	
(b) To prevent the ammeter from overloading (when PQ is shorted or the resistance to be measured is too small)	1A	
	1	

9. (a)	(i)	β decay / beta decay OR ${}^{40}_{19}\text{K} \rightarrow {}^{40}_{20}\text{Ca} + {}^{0}_{-1}\beta$	1A		
				1	
	(ii)	Justified: the penetrating power of β radiation enables it to penetrate the body's organ / skin Or Not justified: the activity is low and is comparable to background radiation / β radiation is largely shielded by the human body	1A		
				1	
(b)	(i)	$\frac{0.45 \times 0.012\%}{40.0} = 1.35 \times 10^{-6}$ (mole)	1A		
				1	
	(ii)	$k = \frac{\ln 2}{1.25 \times 10^9\ (3.16 \times 10^7)} = 1.754803 \times 10^{-17}$ (s^{-1})	1M/1A		i. e. 5.545177×10^{-10} (year^{-1})
		Activity = kN $= 1.754803 \times 10^{-17} \times (1.35 \times 10^{-6} \times 6.02 \times 10^{23})$ $= 14.261284$ (Bq) ≈ 14.3 (Bq)	1A		Accept 14.2 to 14.3 (Bq)
				2	